THE DISCREPANCY OF SYLVESTER–HADAMARD MATRICES IN ODD DIMENSION

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ABSTRACT. Let H_k be the $2^k \times 2^k$ Sylvester–Hadamard matrix and let $\operatorname{disc}(H_k)$ be the minimum of $\|H_k x\|_\infty$ over sign vectors x. It is known that $\operatorname{disc}(H_k) = 2^{k/2}$ for even k, and it has been conjectured that $\operatorname{disc}(H_k) = 2^{(k+1)/2}$ for every odd k. We show that $\operatorname{disc}(H_k) = 2^k - 2\rho(1,k)$, where $\rho(1,k)$ is the covering radius of the first-order Reed–Muller code of length 2^k. Known values and bounds for this covering radius then show that the conjecture holds for $k \le 7$ and fails for every odd $k \ge 9$; in fact $\operatorname{disc}(H_k) \le \frac{7}{8}\, 2^{(k+1)/2}$ for every odd $k \ge 9$, and $\operatorname{disc}(H_9) = 28$ by the determination $\rho(1,9) = 242$ of Wang, Lou and Wang. Determining $\operatorname{disc}(H_k)$ for odd $k \ge 11$ is therefore equivalent to determining $\rho(1,k)$.

1. INTRODUCTION

For a real $m \times n$ matrix A the discrepancy is

$$\operatorname{disc}(A) = \min_{x \in \{-1,1\}^n} \|Ax\|_\infty.$$

Index the rows and columns of the Sylvester–Hadamard matrix H_k by $\mathbb{F}_2^k$, so that $(H_k)_{a,y} = (-1)^{a \cdot y}$. By Parseval every $x \in \{-1,1\}^{2^k}$ satisfies $\|H_k x\|_2^2 = 2^{2k}$, hence $\|H_k x\|_\infty \ge 2^{k/2}$, and $\operatorname{disc}(H_k) = 2^{k/2}$ when k is even. The problem list [1] records the odd case as open and states the following.

Conjecture ([1, Conjecture 13]). For every odd k, $\operatorname{disc}(H_k) = \sqrt{2}\,\sqrt{2^k} = 2^{(k+1)/2}$.

The value $2^{(k+1)/2}$ is what one obtains from H_{k-1} and a bent function, and the question is whether it can be improved. The answer is a statement about Boolean functions that is classical in coding theory [2, 3]. For $f\colon \mathbb{F}_2^k \to \mathbb{F}_2$ let

$$W_f(a) = \sum_{y \in \mathbb{F}_2^k} (-1)^{f(y) + a \cdot y}$$

be its Walsh transform, and let $\mathrm{NL}(f)$ be the Hamming distance from f to the set of affine functions. The covering radius of the first-order Reed–Muller code $RM(1,k)$, the code of affine functions, is $\rho(1,k) = \max_f \mathrm{NL}(f)$.

Theorem 1. *For every $k \ge 1$, $\operatorname{disc}(H_k) = 2^k - 2\rho(1,k)$.*

Corollary 2. (1) $\operatorname{disc}(H_k) = 2^{(k+1)/2}$ *for* $k \in \{1,3,5,7\}$.

(2) $\operatorname{disc}(H_9) \le 28 < 32 = 2^5$.

(3) *For every odd $k \ge 9$, $\operatorname{disc}(H_k) \le \frac{7}{8}\, 2^{(k+1)/2}$. In particular the Conjecture fails for every odd $k \ge 9$.*

(4) *For every odd $k \ge 15$, $\operatorname{disc}(H_k) \le \frac{27}{32}\, 2^{(k+1)/2}$.*

2. Proofs

Proof of Theorem 1. Every $x \in \{-1,1\}^{\mathbb{F}_2^k}$ is $x = (-1)^f$ for a unique $f\colon \mathbb{F}_2^k \to \mathbb{F}_2$, and then $(H_k x)_a = \sum_y (-1)^{a\cdot y + f(y)} = W_f(a)$. Writing $d(f,g)$ for Hamming distance,

$$d(f, a\cdot y) = \tfrac{1}{2}\big(2^k - W_f(a)\big), \qquad d(f, a\cdot y + 1) = \tfrac{1}{2}\big(2^k + W_f(a)\big),$$

so the distance from f to the affine functions is $\mathrm{NL}(f) = \frac{1}{2}\big(2^k - \max_a |W_f(a)|\big)$. Hence

$$\|H_k x\|_\infty = \max_a |W_f(a)| = 2^k - 2\,\mathrm{NL}(f),$$

and minimizing over x is maximizing $\mathrm{NL}(f)$ over f. □

Lemma 3. *Let $f\colon \mathbb{F}_2^k \to \mathbb{F}_2$ and let $b\colon \mathbb{F}_2^m \to \mathbb{F}_2$ be bent (m even). The function $g(y,z) = f(y) + b(z)$ on $\mathbb{F}_2^{k+m}$ satisfies $\max|W_g| = 2^{m/2}\max|W_f|$. Consequently $\mathrm{disc}(H_{k+m}) \le 2^{m/2}\,\mathrm{disc}(H_k)$ for every even $m \ge 0$.*

Proof. The sum defining $W_g(a,c)$ factors as $W_f(a)\,W_b(c)$, and $|W_b(c)| = 2^{m/2}$ for every c because b is bent. Bent functions exist in every even dimension [10]. The second statement follows from Theorem 1, applied with f of maximal nonlinearity. □

Proof of Corollary 2. (1) The covering radii are $\rho(1,1) = 0$ and $\rho(1,3) = 2$ by inspection, $\rho(1,5) = 12$ [4], and $\rho(1,7) = 56$ [5]. Theorem 1 gives $\mathrm{disc}(H_k) = 2, 4, 8, 16$ for $k = 1, 3, 5, 7$, which is $2^{(k+1)/2}$ in each case.

(2) Kavut and Yücel [8] found nine-variable Boolean functions of nonlinearity 242, so $\rho(1,9) \ge 242$ and $\mathrm{disc}(H_9) \le 512 - 484 = 28$. (Nonlinearity 241, found earlier by Kavut, Maitra and Yücel [7], already gives $\mathrm{disc}(H_9) \le 30$.) Wang, Lou and Wang [9] prove that $\rho(1,9) = 242$, so in fact $\mathrm{disc}(H_9) = 28$.

(3) Every odd $k \ge 9$ is $9 + m$ with m even. By Lemma 3 and (2), $\mathrm{disc}(H_k) \le 28 \cdot 2^{m/2} = \frac{28}{32}\,2^{(9+m+1)/2}$.

(4) Patterson and Wiedemann [6] showed $\rho(1,15) \ge 16276$, so $\mathrm{disc}(H_{15}) \le 32768 - 32552 = 216 = \frac{27}{32}\cdot 256$. Lemma 3 carries this to every odd $k \ge 15$. □

3. Remarks

Remark 4. Part (4) uses only results available since 1983, so the Conjecture was false for $k \ge 15$ before it was stated. The first odd dimension in which it fails is $k = 9$.

Remark 5. By Theorem 1 the odd case of the problem in [1] is the problem of the covering radius of $RM(1,k)$ for odd k, open since the 1970s [2] and settled for $k \le 9$ [4, 5, 9]. For odd k the known bounds are

$$2^{k/2} < \mathrm{disc}(H_k) \le 2^{(k+1)/2},$$

where the strict lower bound is the classical fact that bent functions exist only in even dimension [10], and the upper bound is improved by the constructions above for every odd $k \ge 9$. Any new lower bound on $\rho(1,k)$, and any upper bound on $\rho(1,k)$, transfers to $\mathrm{disc}(H_k)$ through Theorem 1 without loss.

Remark 6. The list [1] also states Conjecture 12, $\limsup_{n\to\infty} \sup_{A\in\{\pm1\}^{n\times n}} \mathrm{disc}(A)/\sqrt{n} > 1$, and notes that Conjecture 13 would imply it through H_k, k odd. By Theorem 1, Sylvester–Hadamard matrices witness Conjecture 12 if and only if

$$\limsup_{k \text{ odd}} \big(2^k - 2\rho(1,k)\big)\,2^{-k/2} > 1.$$

Corollary 2 bounds this lim sup by $\frac{27}{32}\sqrt{2} \approx 1.19$; whether it exceeds 1 is a question about the covering radius of $RM(1,k)$ that the present note leaves open.

Remark 7. The repository that accompanies this note contains `check.py`, which verifies Theorem 1 by exhaustive search for $k \leq 4$ and checks the product identity of Lemma 3 on random functions.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [11] and working together inside the Brain framework [13], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [12], the Math Brain repository [11] and the Brain framework repository [13].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

Anthropic Claude AI systems were used to develop the mathematical argument, write and run the verification script, check the cited literature values, and write and prepare this manuscript and its publication files. The author selected the problem and directed the work; the author did not carry out the mathematics or the writing. Responsibility for publication rests with the author. The AI-assisted internal reviews and compilation checks are not external peer review or formal verification.

Data and code availability

The script `check.py` described in Section 3 is in the repository

`github.com/yspennstate/hadamard-discrepancy-odd`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [11].

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